

Study of the luminescent properties of Ce-doped strontium aluminosilicate ($\text{SrAl}_2\text{Si}_2\text{O}_8$) synthesized by the solid-state reaction method

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1. INTRODUCTION

In the last decade, materials such as silicates, aluminates, and aluminosilicates have been studied due to their attractive TL properties for dosimetry applications. Among these, strontium aluminosilicate (SASO), with excellent luminescent properties, has also shown good TL response to gamma and UV radiation (1,2). SASO doped with rare earth elements such as Ce, Tb, and Dy has been synthesized using the solid-state reaction (SSR) method (3,4), which is a simple and low-cost method compared to others, but allows the formation of a stable crystalline structure (matrix) at high temperatures. Therefore, in this work, we synthesized the Cerium-doped SASO by SSR and studied its TL properties under gamma radiation effect.

2. MATERIALS AND METHODS

Ce^{3+} doped SASO phosphor was synthesized by solid-state reaction. Reagents SrCO_3 , Al_2O_3 , SiO_2 , CeO_2 (as a dopant), and H_3BO_3 (as a flux) were used as raw materials.

Initially, the reactants were mixed using a ball mill for 4 hours, obtaining a homogeneous sample. The sample was then placed in a furnace at 1200°C of temperature during three hours and slow cooling. After the synthesis, the sample was ground using an agate mortar and pestle to form a very fine powder. The powder sample was used to perform XRD, while pellets had to be made for the TL measurements.

3. RESULTS

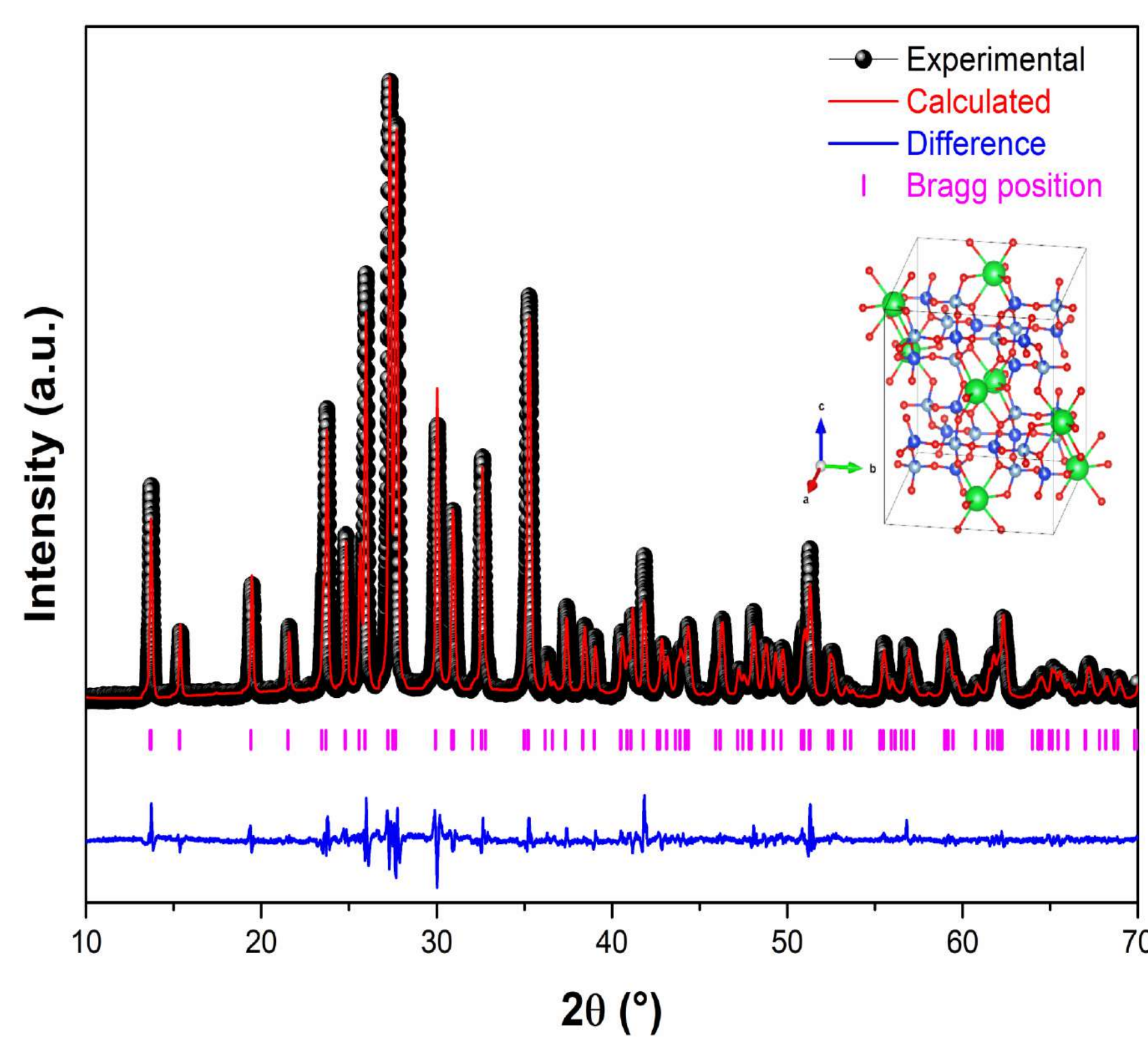


Fig.1 XRD patterns SASO:Ce 0.3%

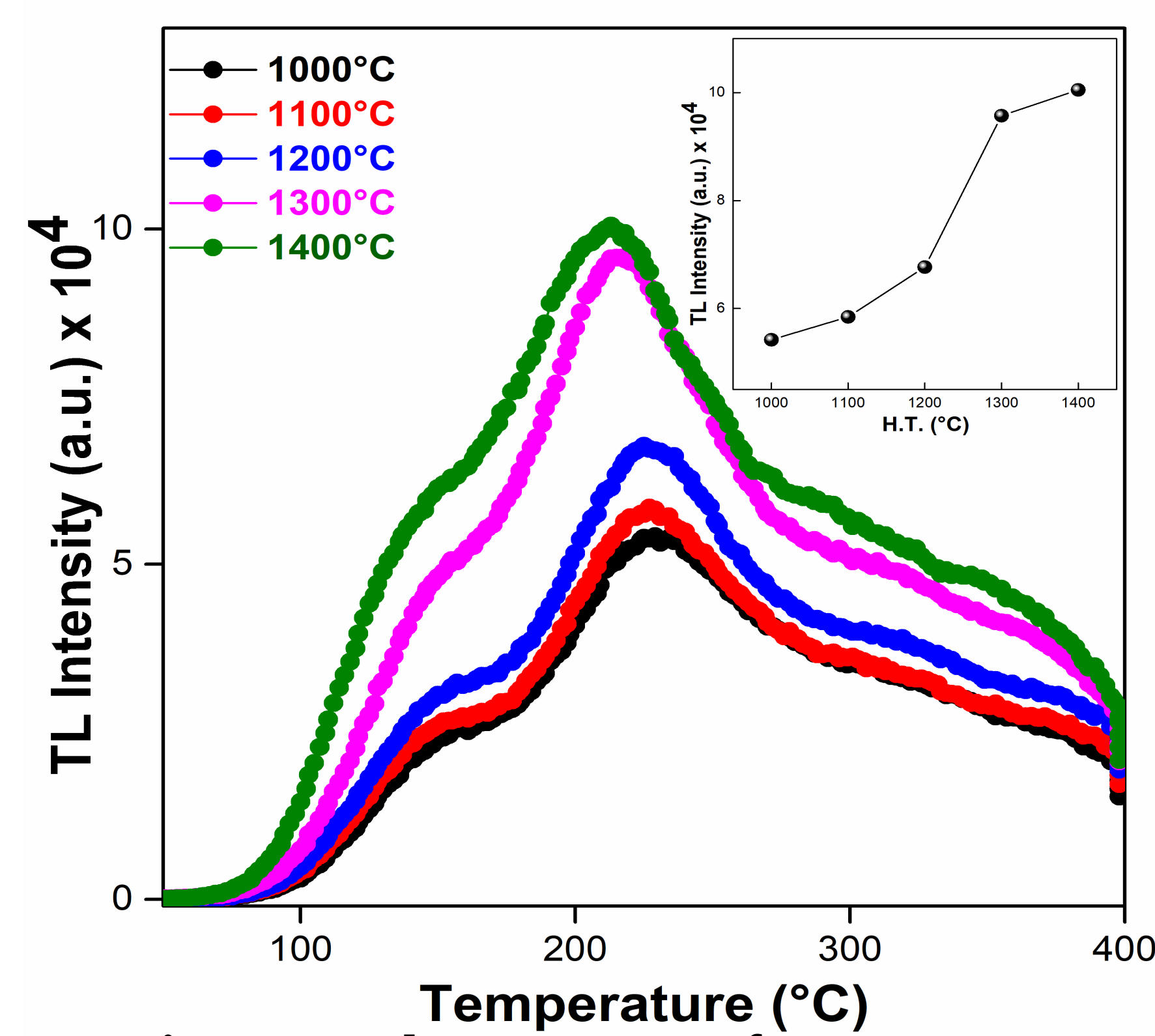


Fig.3 TL glow curves of 0.3%Ce-SASO pellets with different H.T.

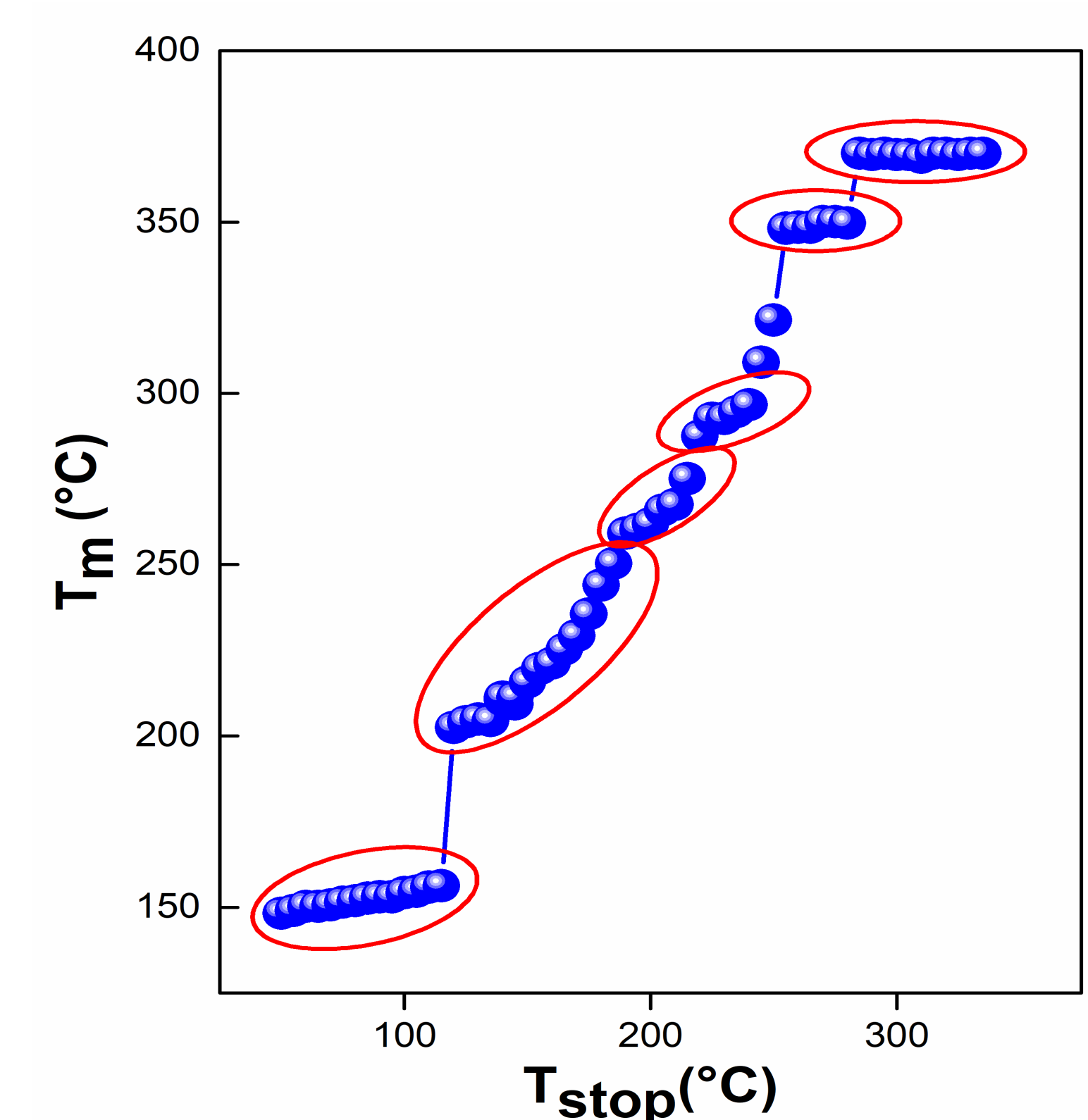


Fig.5 SASO Tm-Tstop:Ce 0.3% with heat treatment of 1300°C .

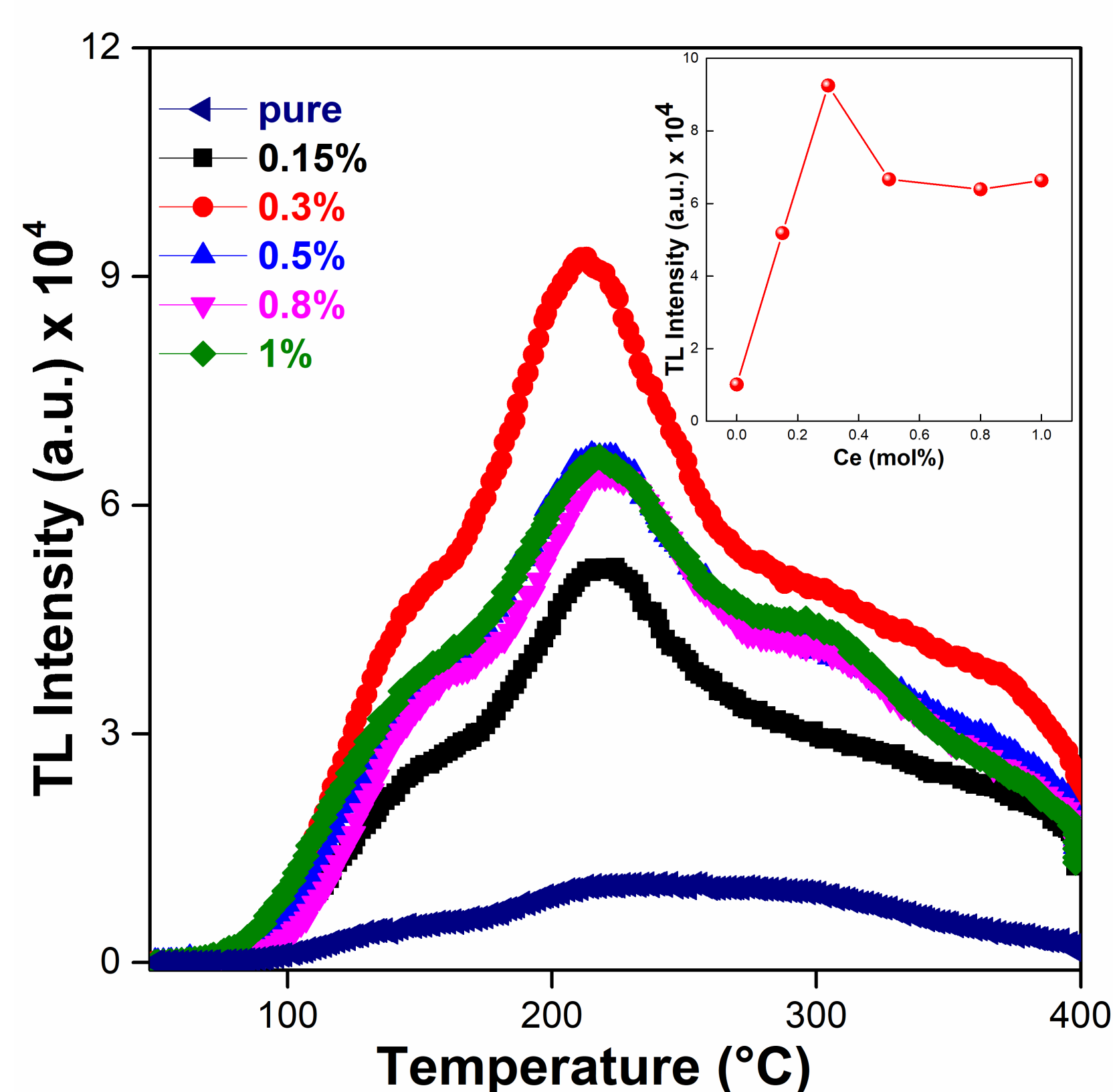


Fig.2 TL glow curves of SASO pellets doped with different % of Ce.

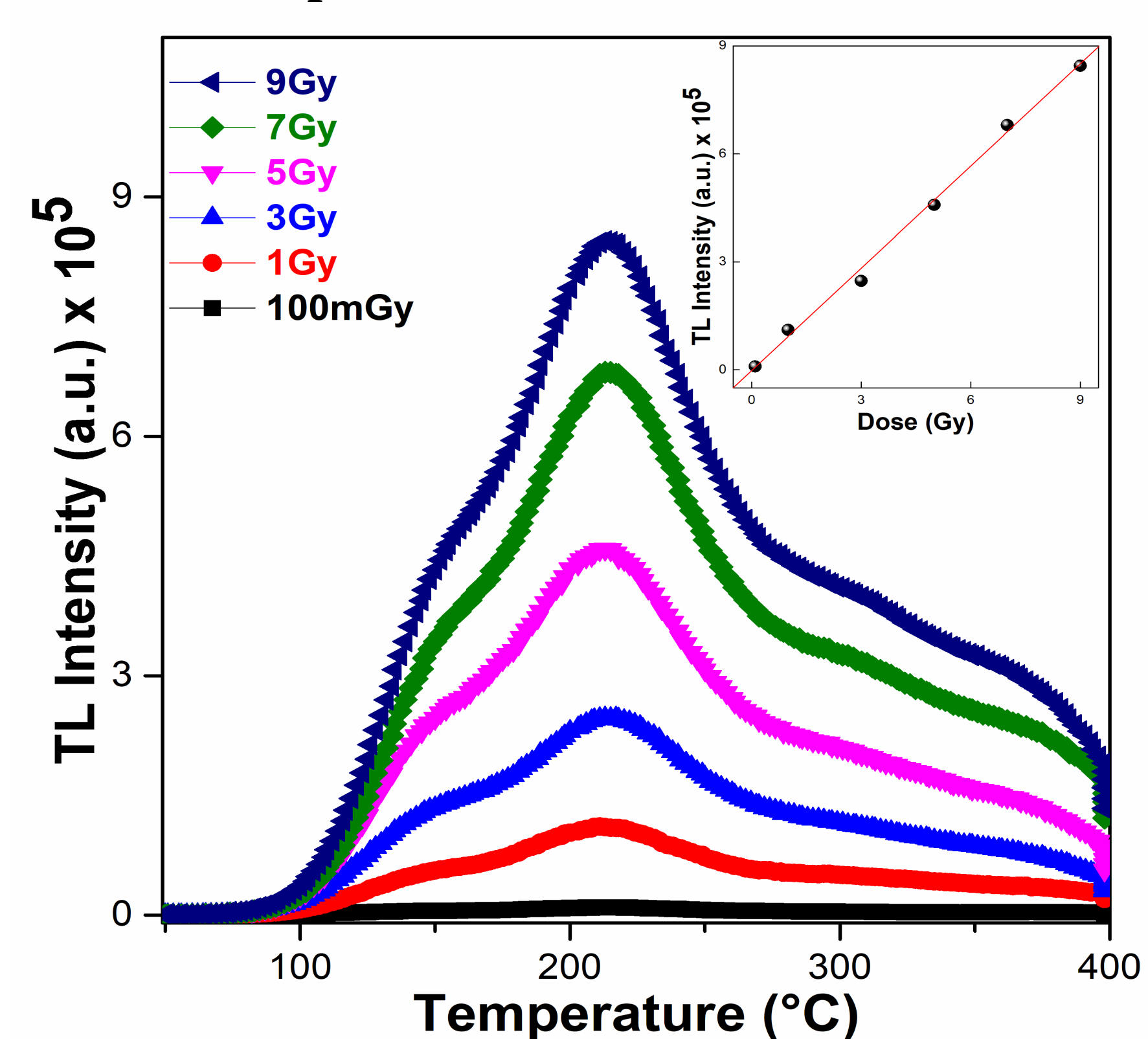


Fig.4 TL curves of SASO:Ce 0.3% with heat treatment of 1300°C .

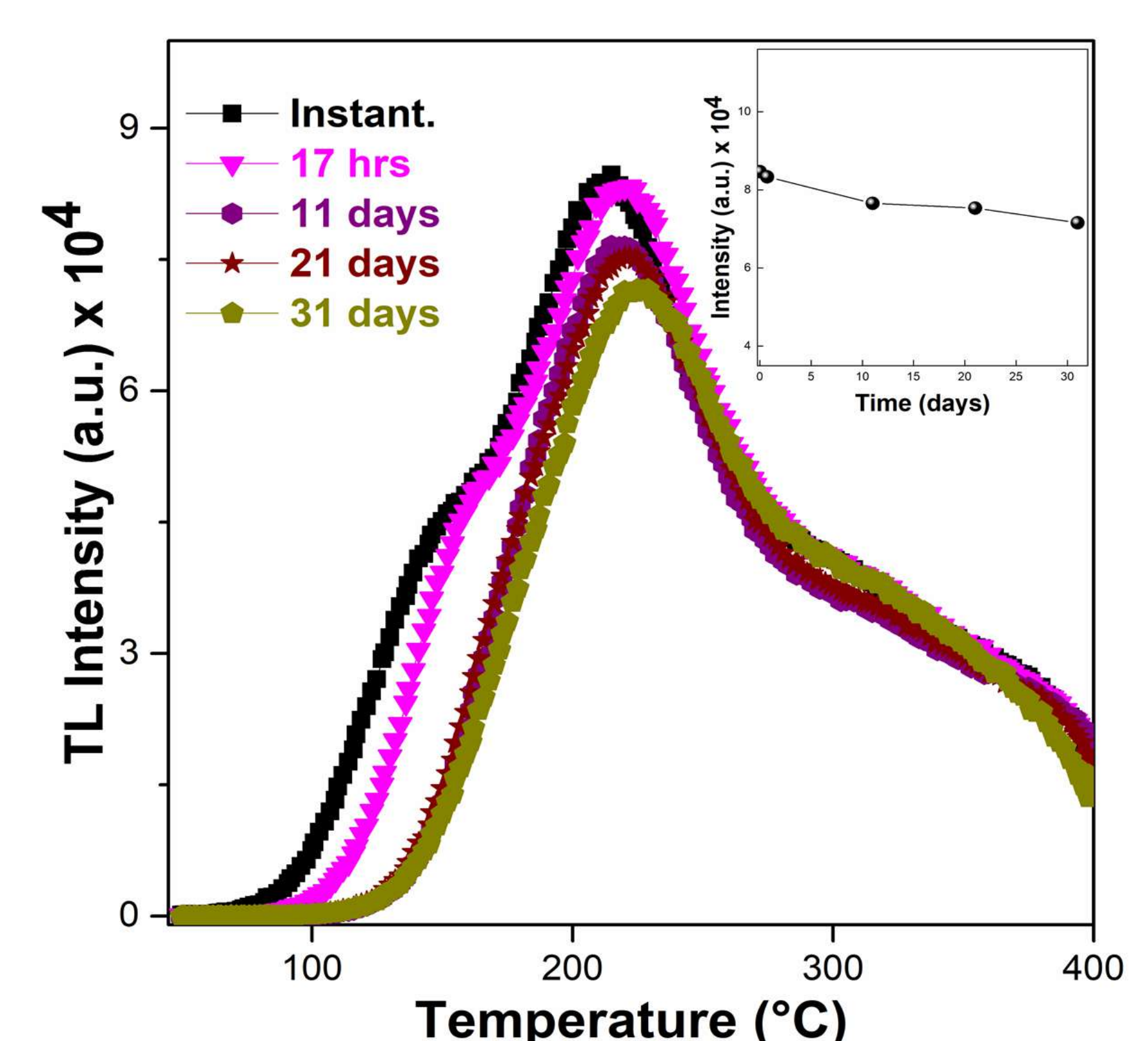


Fig.6 Fading of the TL glow curves of SASO doped with 0.3 mol% Ce.

4. CONCLUSIONS

The Ce-doped SASO samples were synthesized using the solid-state reaction method, through XRD analysis we verified the crystalline structure like a monoclinic phase ($\text{SrAl}_2\text{Si}_2\text{O}_8$). The sample that present the best thermoluminescent response was the one with a 0.3% mol concentration of cerium. In addition, when this sample was subjected to different thermal treatments (TT), it was found that at higher temperatures the TL emission curves increased their intensity showing the most TL intensity at 1300°C of TT. The 0.3% Ce-doped SASO shows linear behavior when increase the dose of gamma radiation from 100 mGy to 9 Gy. The Tm-Tstop method revealed that the TL curve is composed of six internal peaks that represent traps in the structure. The fading of the TL intensity after 31 days was approximately 14%.

References

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